

PLAS-Net: Lightweight Instance Segmentation of Beach Litter in Unmanned Aerial Vehicle Imagery

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Abstract—Unmanned aerial vehicles can survey more beach area than field teams can cover efficiently. Natural backgrounds and partly buried objects, however, complicate automated segmentation. We present the Pixel-level Litter Area Segmentor (PLAS-Net), a lightweight model that produces an individual mask for each item in high-resolution aerial imagery. PLAS-Net adds three feature-fusion and attention modules to a YOLO26n-seg baseline. A survey at Aow Luek Bay, Thailand, produced a 0.17-cm-per-pixel orthomosaic and 1,300 non-overlapping image tiles. On the 395-image evaluation split, PLAS-Net achieved 69.7 percent mask precision and 58.7 percent mean average precision at an intersection-over-union threshold of 0.5. Mean average precision averaged over thresholds from 0.5 to 0.95 was 46.8 percent. These three metrics exceeded the baseline by 16.5, 5.1, and 2.5 percentage points. The parameter count increased from 2.69 to 2.73 million. Scene-wide inference also produced size-frequency and footprint summaries from the predicted masks.

Index Terms—beach litter, instance segmentation, marine debris, remote sensing, unmanned aerial vehicle

I. INTRODUCTION

Marine debris persists in coastal environments and affects ecological processes, navigation, fisheries, and tourism [1]. Beach surveys therefore require repeatable observations over large areas. Conventional transect surveys provide direct evidence, but field effort limits their coverage and frequency. Unmanned aerial vehicles (UAVs) reduce acquisition time and provide georeferenced imagery at centimeter or sub-centimeter resolution [2], [3].

Recent methods use object detection, semantic segmentation, or instance segmentation to analyze UAV imagery. Object detectors count items, but their rectangular boxes include background pixels around irregular litter. Semantic segmentation identifies covered pixels but cannot separate adjacent objects. Instance segmentation assigns a separate mask to each item. These masks preserve object-level position, size, and morphology. Previous litter studies have used instance segmentation [4], pixel-level classification [5], and underwater or aerial-aquatic imaging with restoration and detection networks [12], [13].

Natural beaches remain a difficult visual setting. Sand, shells, vegetation, and wrack produce texture that resembles weathered debris. Partial burial interrupts object boundaries,

and fishing lines or nets may span several feature scales. These conditions require background suppression, long-range spatial context, and cross-scale fusion within a compact model.

Mask quality affects environmental summaries as well as recognition accuracy. False detections increase both item counts and estimated area. Fragmented masks alter the apparent size distribution, even when the class label is correct. Evaluation must therefore cover precision, recall, mask accuracy, model size, and throughput.

We developed the Pixel-level Litter Area Segmentor (PLAS-Net) from YOLO26n-seg [7]. The model contains three modifications. C3k2 with Dynamic Feature Fusion (C3kDFF) weights channel and spatial features during fusion. Cross-Stage Partial Spatial Attention (C2PSA) is combined with Context Anchor Attention to form CPCAA. This module links weak or occluded regions along directional receptive fields. Dynamic Multi-Scale Sequence Fusion (DMSSF) aligns features across spatial scales to retain small and elongated objects. We compare PLAS-Net with eleven segmentation models and conduct a cumulative ablation. We then use its masks to estimate object area across the survey.

II. MATERIALS AND METHODS

A. Survey and Dataset

Aow Luek Bay is a semi-enclosed pocket beach on the southeast coast of Koh Tao in the Gulf of Thailand. Headlands and seasonal onshore transport favor local debris retention [6]. Red-green-blue (RGB) imagery was collected in February 2024 with a DJI Mini 3 at an altitude of 5 m. Forward overlap was at least 75 percent, and lateral overlap was at least 65 percent. Agisoft Metashape was used for structure-from-motion (SfM) reconstruction. The resulting orthomosaic had a ground sampling distance (GSD) of 0.17 cm/pixel (Fig. 1).

The orthomosaic was divided into 1,300 non-overlapping 512×512 tiles. We used 905 tiles for model fitting and 395 tiles for evaluation. The evaluation split contained 852 annotated instances in 13 litter categories. Two annotators delineated visible object boundaries with polygon masks. Partly buried items were traced along their visible perimeter. Thin items were assigned a minimum polygon width of two pixels. A second annotator cross-checked a random 15 percent of the tiles. Boundary disagreements greater than 10 percent intersection over union (IoU) were resolved by consensus.

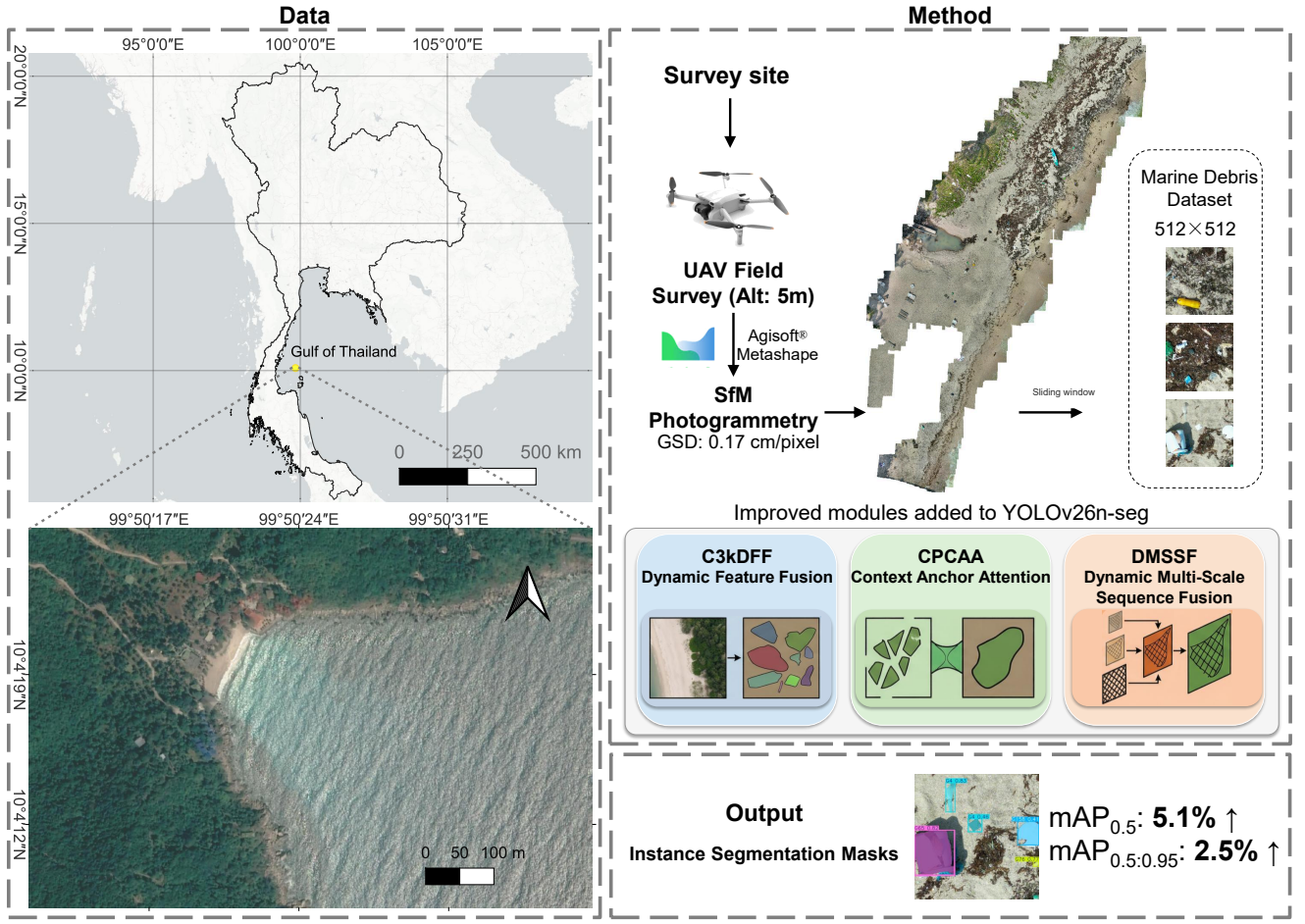


Fig. 1. Survey and analysis workflow. Unmanned aerial vehicle (UAV) imagery from Aow Luek Bay was reconstructed into a 0.17-cm-per-pixel orthomosaic and divided into 512×512 tiles. PLAS-Net adds C3kDFF, CPCAA, and DMSSF to YOLO26n-seg. The model returns one instance mask for each detected item.

B. PLAS-Net

PLAS-Net retains the YOLO26n-seg backbone, feature pyramid, and segmentation head. The modifications affect feature extraction and fusion. Each input image is resized to 640×640 pixels. For every instance, the model returns a class, confidence score, bounding box, and binary mask.

1) *Dynamic Feature Fusion*: C3kDFF replaces static residual fusion at selected backbone and neck stages. Let X and S denote the main and residual features. Their concatenation $F = [X; S]$ is calibrated by a channel gate α and a spatial gate β :

$$\alpha = \sigma(C_{1 \times 1}(\text{GAP}(F))), \quad (1)$$

$$\beta = \sigma(C_{1 \times 1}(X) + C_{1 \times 1}(S)), \quad (2)$$

$$Y = \beta \odot C_{1 \times 1}(\alpha \odot F), \quad (3)$$

where $C_{a \times b}$ denotes a convolution with an $a \times b$ kernel. The operators GAP and σ denote global average pooling and sigmoid activation, respectively. The symbol $\odot$ denotes element-wise multiplication, and Y is the fused output. The

two gates reduce contributions from irrelevant channels and spatial regions before residual fusion.

2) *Context Anchor Attention*: CPCAA replaces the baseline attention block with context anchor attention [8]. Average pooling first aggregates local context. Pointwise compression is followed by horizontal and vertical depth-wise strip convolutions. The attention map and output are

$$Z = C_{k \times 1}(C_{1 \times k}(C_{1 \times 1}(\text{AvgPool}(X)))), \quad (4)$$

$$A(X) = \sigma(C_{1 \times 1}(Z)), \quad Y = A(X) \odot X. \quad (5)$$

Here, AvgPool denotes average pooling, k is the strip-kernel length, Z is the context feature, and $A(X)$ is the attention map. The directional receptive field links separated regions along slender or partly occluded objects. This operation avoids full self-attention.

3) *Dynamic Multi-Scale Sequence Fusion*: DMSSF receives the P3, P4, and P5 levels of the feature pyramid. Pointwise convolutions first align their channel dimensions. DySample then resamples P4 and P5 to the P3 resolution [9].

The module stacks the aligned maps along a scale dimension and computes

$$Y = \text{MaxPool3D}(\phi(\text{BN}(C_{3D}([P_3; P'_4; P'_5])))), \quad (6)$$

Here, P'_4 and P'_5 are the resampled feature maps; the semicolons indicate concatenation along the scale dimension. The operators C_{3D} , MaxPool3D , BN , and ϕ denote three-dimensional convolution, three-dimensional max pooling, batch normalization, and LeakyReLU, respectively. The scale-sequence design follows compact instance-segmentation networks [10]. Dynamic sampling reduces the loss of local detail during alignment.

C. Training and Evaluation

All models were trained for 200 epochs in PyTorch 2.7.0 with CUDA 12.8 on an NVIDIA RTX 5090 graphics processing unit (GPU). The batch size was 32. Stochastic gradient descent used an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. Training used seed 0 with deterministic execution. Augmentation included hue-saturation-value (HSV) variation and translation up to 0.1. Images were also scaled by up to 0.5 and flipped horizontally with a probability of 0.5. Mosaic augmentation was disabled for the final ten epochs.

The Ultralytics evaluation pipeline computed mask precision, mask recall, mAP_{50} , and $\text{mAP}_{50:95}$. The mAP_{50} metric is mean average precision at an IoU threshold of 0.5. The $\text{mAP}_{50:95}$ metric averages mean average precision over IoU thresholds from 0.5 to 0.95. Throughput was measured at batch size one with 640×640 32-bit floating-point (FP32) tensors. All models were fused before timing. For each model, 200 warm-up iterations preceded 1,000 timed forward passes. CUDA synchronization was applied during timing. The reported frames per second (FPS) excludes image decoding, preprocessing, and postprocessing.

III. RESULTS

A. Comparison with Segmentation Models

Table I compares PLAS-Net with eleven instance-segmentation models on the same data split and at the same input size. PLAS-Net achieved the highest mask precision, mAP_{50} , and $\text{mAP}_{50:95}$. Relative to YOLO26n-seg, precision increased from 53.2 to 69.7 percent. The mAP_{50} value increased from 53.6 to 58.7 percent, and $\text{mAP}_{50:95}$ increased from 44.3 to 46.8 percent. The parameter count increased by 0.04 million. Recall decreased from 46.7 to 44.4 percent.

Model size did not determine mask accuracy. RT-DETR-L used 30.81 million parameters but reached 50.8 percent mAP_{50} . Among the compact models, YOLOv11n-seg achieved the highest comparison values: 52.6 percent mAP_{50} and 43.9 percent $\text{mAP}_{50:95}$. PLAS-Net exceeded these values by 6.1 and 2.9 percentage points, respectively, with 2.73 million parameters. PLAS-Net processed 145 frames per second on the RTX 5090 GPU.

B. Cumulative Ablation and Qualitative Results

The cumulative ablation in Table II produced a monotonic increase in mAP_{50} and $\text{mAP}_{50:95}$. C3kDFF raised mAP_{50} by 1.9 points and precision by 10.3 points. Adding CPCA increased mAP_{50} by a further 2.3 points and raised recall to 46.8 percent. DMSSF produced the final 0.9-point gain in both mAP measures. The complete model used 1.5 percent more parameters than the baseline.

Figure 2 shows three difficult scenes. In the low-contrast scene, the baseline omitted several annotated objects. Successive modules recovered some of these objects and produced masks closer to the annotated boundaries. In the dense-vegetation scene, PLAS-Net detected several annotated fragments missed by the baseline. PLAS-Net also produced predictions without matching ground-truth masks. These false positives occurred mainly near vegetation and other high-contrast background textures. In the wrack-sand scene, PLAS-Net recovered additional small targets but still missed annotated objects and produced an unmatched prediction. The examples therefore show both improved object recovery and residual confusion with natural beach material.

C. Mask-Derived Environmental Summaries

Projected area was calculated for each instance as $A_i = n_i \text{GSD}^2$, where n_i is the number of mask pixels. Figure 3 presents two summaries from PLAS-Net inference over the full orthomosaic. Above the 625-mm² macro-litter threshold, the intertidal distribution had a power-law slope of -2.14 ($R^2 = 0.96$). The backshore slope was -1.88 ($R^2 = 0.99$). Both fits followed the size-frequency formulation in [11].

Scene-wide inference returned 2,957 instances. Fragments represented 58.1 percent of the item count but 37.5 percent of the cumulative mask area. Domestic items represented 41.3 percent of the count and 59.8 percent of the area. Fishing items represented 0.6 percent of the count and 2.7 percent of the area. Count and footprint rankings therefore differed among the three source groups.

IV. DISCUSSION

The three modules affected different stages of the segmentation pipeline. C3kDFF produced the largest precision gain, consistent with stronger background filtering. CPCA produced the largest incremental mAP_{50} gain and increased recall, consistent with stronger spatial support for weak object regions. DMSSF improved both mAP measures, and the complete model contained 2.73 million parameters. The cumulative ablation measures each stage of the final design.

Parameter count did not predict either speed or accuracy. Several models with fewer than 4 million parameters exceeded 180 frames per second, but their mask accuracy varied. RT-DETR-L-seg used more than eleven times as many parameters as PLAS-Net and had lower mask mAP . PLAS-Net was slower than YOLO26n-seg but achieved higher precision and both mAP measures. Its throughput supports offline screening of tiled orthomosaics.

TABLE I

INSTANCE-SEGMENTATION PERFORMANCE ON THE 395-IMAGE EVALUATION SPLIT. ALL ACCURACY VALUES ARE PERCENTAGES. MODEL SIZE IS REPORTED IN MEGABYTES (MB). THROUGHPUT IS MODEL-FORWARD SPEED AT BATCH SIZE ONE WITH FP32 INPUTS ON AN RTX 5090 GPU. BEST VALUES ARE IN BOLD.

Model	Precision	Recall	mAP ₅₀	mAP _{50:95}	Params (million)	Size (MB)	FPS
RT-DETR-L-seg	61.2	45.9	50.8	41.8	30.81	63.8	143
GhostNet-seg	42.4	35.4	35.7	27.1	3.01	6.5	183
Mamba-YOLO-T-seg	27.2	58.7	49.4	39.6	6.23	12.8	63
YOLOv6s-seg	50.8	28.6	38.5	29.1	16.75	33.8	291
YOLOv8n-seg	45.4	38.0	38.6	30.3	3.26	6.8	289
YOLOv9s-seg	40.9	49.9	50.5	41.3	8.52	18.0	107
YOLOv10n-seg	46.0	41.3	49.7	41.2	2.52	5.4	240
YOLOv11n-seg	52.1	44.3	52.6	43.9	2.84	6.0	232
YOLOv12n-seg	57.0	34.1	51.3	41.8	2.81	6.1	169
YOLOv13n-seg	33.9	50.1	50.4	42.0	2.72	6.0	117
YOLO26n-seg	53.2	46.7	53.6	44.3	2.69	6.6	193
PLAS-Net	69.7	44.4	58.7	46.8	2.73	6.7	145

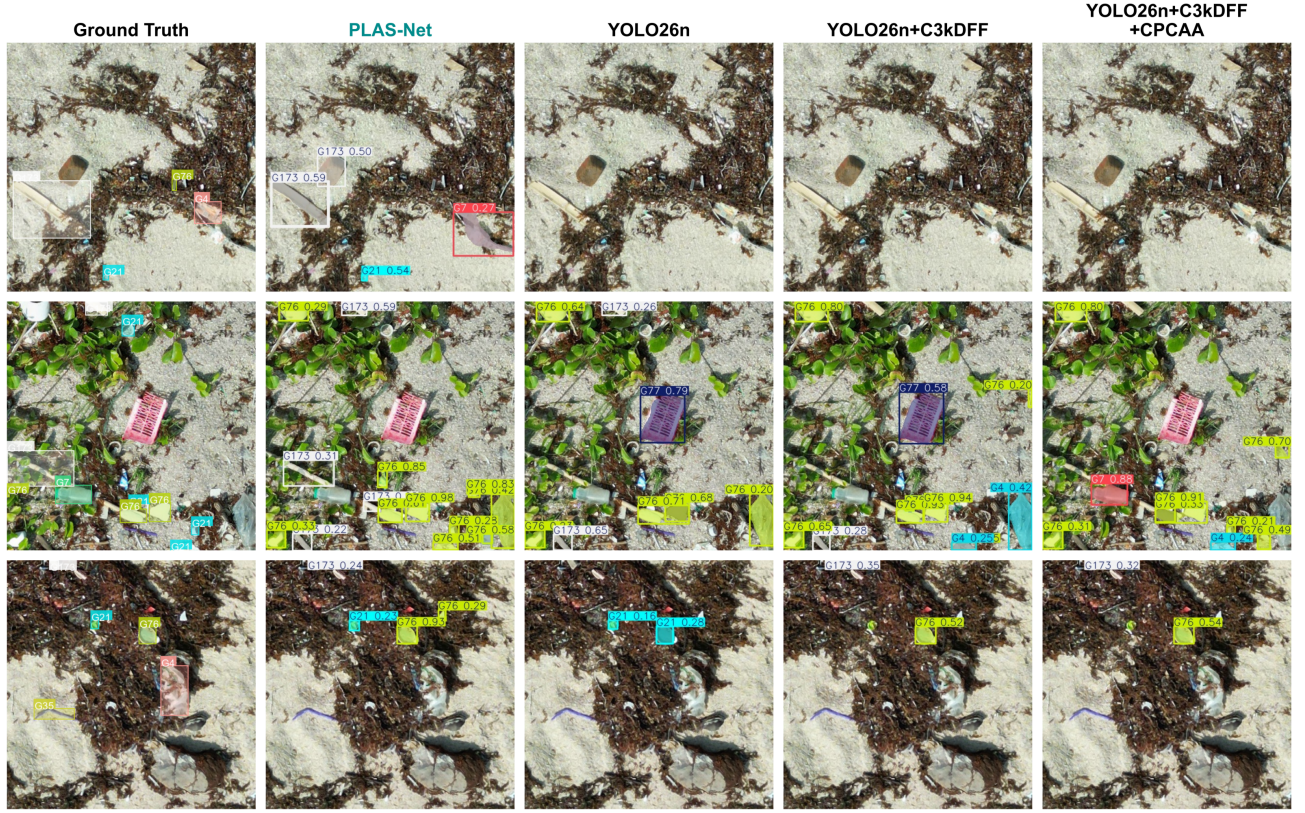


Fig. 2. Cumulative qualitative comparison. Columns show the ground truth, PLAS-Net, YOLO26n-seg, YOLO26n-seg with C3kDFF, and YOLO26n-seg with C3kDFF and CPCAA. Rows show low-contrast litter on sand, dense vegetation with closely spaced litter, and small targets along a wrack-sand boundary. Labels give the litter-category code and prediction confidence.

PLAS-Net favored precision over recall. Higher precision limits the inflation of projected area caused by background detections. The lower recall means that faint or heavily buried items can still be missed. The qualitative cases clarify this trade-off. Contextual fusion recovered several low-contrast objects. Dense vegetation and high-contrast natural textures still produced unmatched or overlapping predictions.

Scene-wide inference converted each predicted mask into an item count and a projected area. Differences between

count and area shares therefore reflect mask geometry rather than changes in classification. The fitted size-frequency slopes provide a second summary of the observed mask distribution. These slopes can be compared between beach zones. The survey provides a site-specific example from one tropical pocket beach. Additional surveys could test the workflow across coastal textures, flight heights, and seasons.

The two environmental summaries answer different questions. Counts describe how often an item type occurs. Cumu-

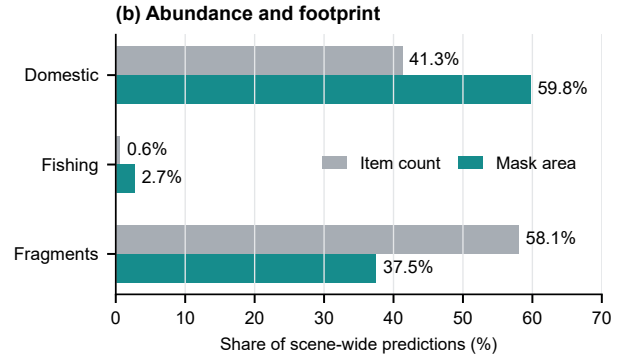
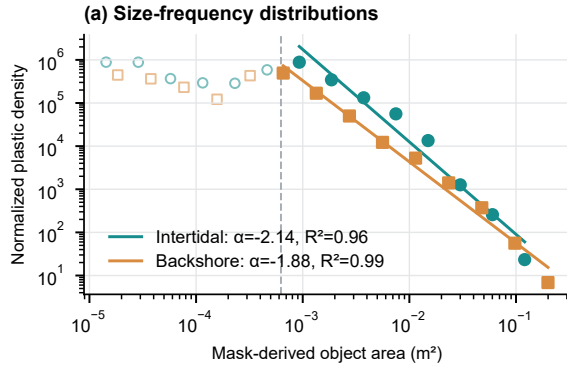


Fig. 3. Mask-derived environmental summaries from scene-wide PLAS-Net inference. (a) Size-frequency distributions and macro-range power-law fits for intertidal and backshore zones. Open symbols fall below the 625-mm² macro threshold and are excluded from the fits. (b) Item-count and cumulative-mask-area shares for three functional source groups.

TABLE II
CUMULATIVE ABLATION OF THE THREE PLAS-NET MODULES. P AND R
DENOTE MASK PRECISION AND MASK RECALL, RESPECTIVELY.
PARAMETER COUNTS ARE IN MILLIONS.

Configuration	P	R	mAP ₅₀	mAP _{50:95}	Params
YOLO26n-seg	53.2	46.7	53.6	44.3	2.69
+ C3kDFF	63.5	42.4	55.5	45.1	2.67
+ CPCAA	65.4	46.8	57.8	45.9	2.66
+ DMSSF	69.7	44.4	58.7	46.8	2.73

lative mask area describes its visible projected footprint in the orthomosaic. A single summary would hide the dominance of fragments by count and domestic items by area. Each value remains linked to one georeferenced mask. Analysts can therefore aggregate results by beach zone or review individual predictions during quality control.

V. CONCLUSION

PLAS-Net combines dynamic feature fusion, directional context attention, and multi-scale sequence fusion to segment beach litter in unmanned aerial vehicle imagery. On the 395-image evaluation split, the model achieved 58.7 percent mAP₅₀ and 46.8 percent mAP_{50:95} with 2.73 million parameters. These values exceeded the YOLO26n-seg baseline by 5.1 and 2.5 percentage points, respectively. PLAS-Net also generated scene-wide size-frequency and footprint summaries from its object masks. These summaries add projected area to conventional item counts.

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